

# Gun Ahn

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## EDUCATION

- **Massachusetts Institute of Technology** Cambridge, MA, USA  
*Ph.D. in Brain and Cognitive Science* Sep. 2024 -
- **Seoul National University** Seoul, South Korea  
*M.S. in Interdisciplinary Program for Bioengineering* Sep. 2021 - June. 2024
- **Seoul National University** Seoul, South Korea  
*B.S. of Materials Science and Engineering* Mar. 2014 - Feb. 2021 (Mar. 2016 - Mar. 2018: Military Service)  
*Combined minor: Brain-Mind-Behavior* Understanding the interplay between the brain and its influence on mind and behavior.

## RESEARCH EXPERIENCE

- **PhD Student (PI: John Gabrieli)** Cambridge, MA, USA  
*Gabrieli Lab, Brain and Cognitive Sciences, Massachusetts Institute of Technology* Oct. 2024 - Present
  - **Predictive Precision Medicine Model for Depression Treatment Personalization:** Designing a machine learning model combining fMRI data and psychosocial variables to predict personalized, evidence-based treatment plans for depression, leveraging historical outcomes to optimize mental health care
- **Visiting Student (PI: Ann M Graybiel)** Cambridge, MA, USA  
*Graybiel Lab, Brain and Cognitive Sciences, Massachusetts Institute of Technology* Dec. 2022 - Aug. 2024
  - **Researched astrocyte functions in relation to dopamine using in-vivo two-color fiber photometry and calcium imaging techniques in both the dorsolateral and dorsomedial striatum regions, focusing on their role in engagement:** Developed a computational pipeline for analyzing task-related behaviors using unsupervised clustering, engagement using HMM, activity of striatal astrocytes, and dopamine release
  - **Investigated the role of the direct dopamine receptor D1 (SD1) and indirect D2 expressing (SD2) neurons in Striosomes:** Utilized reinforcement learning methods to compute action values, decode them from recorded neuronal and dopamine signals, and identify functional relationships between SD1, and SD2 neuron types, and dopamine release.
- **Researcher (PI: Hyung Jin Yoon)** Seoul, South Korea  
*Clinical Big Data Lab, Biomedical Engineering, Seoul National University* Sep. 2021 - Dec.2022
  - **Predicted health outcomes with electrocardiogram signal:** Collected data from the hospital server and predicted mortality, age, type 2 diabetes, and electrolyte abnormalities with a 12-lead electrocardiogram.
  - **Developed generative AI model for transferring non-contrast to contrast CT abdomen images.:** Developed ML model to diagnose renal cell cancer in non-contrast CT abdomen images.
- **ML Developer(Supervisor : Du Hyun Ro)** Seoul, South Korea  
*Seoul National University Hospital Medical Research Center for Innovation* Nov. 2020 - Dec. 2022
  - **Developed generative AI model for medical image synthesis of knee radiographs images:** Pre-processed X-ray knee images and developed GAN, KL-grade classification model for Osteoarthritis knees.
  - **Developed software tool that shows arthritis progression by controlling latent space:** Found the axis in latent space that represents arthritis progression.

## PUBLICATIONS

(\*) EQUAL CONTRIBUTION, (†) CORRESPONDING AUTHOR

- Krishnan, S.(\*), **Ahn, G.(\*)**(†), and Kim, J. 2026. *Beyond the Flag: Clinical Framing Closes the Moderation Gap in Suicide Risk Measurement*. Proceedings of the 5th Workshop on NLP for Positive Impact (NLP4PI), EMNLP 2026. [Link to : Paper]
- **Ahn, G.**, Li, C., Liang, A., Choi W., Ahn, S., Roberts, C., and Gabrieli, J. D. E. 2026. *Large Language Model Few-Shot Learning for Predicting Individual Treatment Response to Smartphone-Based Mindfulness in Autistic Adults With Anxiety: Secondary Analysis of a Randomized Controlled Trial*. Journal of Medical Internet Research Artificial Intelligence. [Link to : Code , Paper]
- Lazaridis, I., **Ahn, G.**, Hirokane, K., Choi W., and Graybiel, A. M. 2024. *Striatal Astrocytes Influence Dopamine Dynamics and Behavioral State Transitions*, Nature Communications (under review) [Link to preprint: Paper]
- Lazaridis, I., Crittenden, J. R., **Ahn, G.**, Hirokane, K., Yoshida, T., Mahar, A., ... and Graybiel, A. M. 2024. *Striosomes control dopamine via dual pathways paralleling canonical basal ganglia circuits*, Current Biology, 34, 5263-5283.e8. [Link to : Paper , MIT News]
- **Ahn, G.(\*)**, Kim, B.(\*), Kim, K.K., Kim, H., Lee, E., Ahn, W.Y., Kim, J.W. and Cha, J., 2021. *Identifying Prepubertal Children with Risk for Suicide Using Deep Neural Network Trained on Multimodal Brain Imaging*. In: Shaban-Nejad, A., Michalowski, M., Bianco, S. (eds.) AI for Disease Surveillance and Pandemic Intelligence. W3PHAI 2021. Studies in Computational Intelligence, vol 1013. Springer, Cham. [Link to : Code , Paper]

- **Ahn, G.(\*)**, Choi, B.S.(\*), Ko, S., Jo, C., Han, H.S., Lee, M.C. and Ro, D.H., 2022. *High-Resolution Knee Plain Radiography Image Synthesis Using Style Generative Adversarial Network Adaptive Discriminator Augmentation*. Journal of Orthopaedic Research®. [Link to : Code , Paper]
- Kim, B.G.(\*), **Ahn, G.(\*)**, Kim, S.(\*), Kim, K., Kim, H., Ahn, W.Y., Kim, J.W. and Cha, J., 2021. *Identifying Prepubertal Children with Risk for Suicide Using Deep Neural Network Trained on Multimodal Brain Imaging-Derived Phenotypes*. medRxiv. [Link to : Code , Paper]
- Kim, K., Joo, Y.Y., **Ahn, G.**, Wang, H.H., Moon, S.Y., Kim, H., Ahn, W.Y. and Cha, J., 2022. *The sexual brain, genes, and cognition: A machine-predicted brain sex score explains individual differences in cognitive intelligence and genetic influence in young children.*, Human Brain Mapping. [Link to : Code, Paper]
- Kim, J., Yang, H., Kim, S.H., Kim, S., Lee, J., Ryu, J., Kim, K., Kim, Z., **Ahn, G.**, Kwon, D., Yoon, H., 2024. *Deep learning-based long-term risk evaluation of incident type 2 diabetes using electrocardiogram in a non-diabetic population: a retrospective, multicentre study*, eClinicalMedicine. [Link to : Paper]

## ENTREPRENEURSHIP

- **Founder** Cambridge, MA, USA  
*Wondi.ai, adolescent AI safety* 2025 - Present
  - **Risk-detection API for adolescent AI companions:** Detects suicidal ideation and self-harm risk in conversation, scored on a graded clinical severity scale grounded in the Columbia Suicide Severity Rating Scale rather than a binary policy flag.
  - **Non-dilutive funding of up to \$70,000:** From MIT Sandbox, the MIT Researcher-to-Entrepreneur (R2E) grant, and Google for Startups.
- **Founder** Seoul, South Korea  
*STEM vision scholarship* Aug. 2020 - Present
  - **Initial capital contribution:** a total of \$20,000+ so far
  - **Scholarship Vision Settings:** Giving scholarship to high school student prospective engineering leaders, who are under-served and have the most potential, 1K\$ per person
  - **Leader of 30+ STEM members for book publication:** *Wanna be an Engineer*, 20,000+ copies sold
  - **Offered Vision Mentoring:** for 1000+ high school students
- **Co-founder** Boston, USA / Seoul, South Korea  
*AI Techplay, Artificial Intelligence Education Institute* Jan. 2021 - Aug. 2022
  - **Co-found a non-profit organization AI Techplay:** Aiming to foster the next-generation engineers through project-based learning supported by MIT Lincoln Lab and SNU Engineering College.
  - **Launched and organized 1st Autonomous Race car AI Coding Competition:** Serve as an author of first autonomous car control practice book(*AI Python Coding in Easy Words with MIT Autonomous Race Car*)

## PUBLICATIONS (NON-ACADEMIC)

- **Ahn, G.** (2020). "The happiest country in the world, Finland" HarmonyBook (ISBN: 9791189930493): In my quest for understanding happiness, I spent 14 months in Finland. The book reveals how Finland became the happiest country in the world for six consecutive years in the World Happiness Report.
- **Ahn, G.** et al.. (2021). "Wanna be an Engineer". Megastduybooks. (ISBN: 9791129708168): The book provides a glimpse into the professional lives of STEM experts and aims to communicate with the public
- Kim, C., Kim.Y, Kim, J, **Ahn, G.** et al.. (2022). "AI Python Coding in Easy Words with Autonomous Race Car" Hongreung. (ISBN:9791156000488) : The book is a learning resource for the computer programming language Python. It combines appropriate Python content with insights from artificial intelligence experts, discussing the future implications.
- **Ahn, G.** et al.. (2023). "Digital Immigration". in "Living Ideas of the Future Observer" Porche (ISBN: 9791192730905): The book is a Science Fiction novel that imagines a future where we overcome the limitations of human mortality by uploading our neuronal circuitry to a server, enabling us to live forever in a digital utopia

## POSTER PRESENTATION

- **Ahn, G.**, Gabrieli, J. *Predicting Suicidal Attempts Using Psychosocial Data: A Foundational model Study from the ABCD Cohort* Society for Neuroscience (SFN) 2025, SanDiego
- **Ahn, G.(\*)**, Choi, W.(\*), Kim, Z.(\*) Kim, D., Ryu, J.(†), Yoon, H(†) *Advances in Renal Cancer CT Imaging: GAN-Enhanced Contrast Visualization and U-Net3+ Based Tumor Detection Techniques*. Korean Society of Artificial Intelligence in Medicine (KoSAIM) 2024 (26, Oct. 2024)
- Lazaridis, I., **Ahn, G.**, Hueske, E., Matsushima, A, Sur, M., Graybiel, A. *Exploitation / Exploration balance: From Striosomes to Dopamine and Astrocytes*. Neuroscience 2023 (11-15, Nov. 2023)

- Lazaridis, I., **Ahn, G.**, Hueske, E., Matsushima, A, Sur, M., Graybiel, A. *Striatal astrocytes in dopamine regulation: A new perspective on behavioral state transitions and interaction with striosomes*. International Basal Ganglia Society (IBAGS)(13-16, June. 2023)
- **Ahn, G.**, Lazaridis, I. Andrews, E. Dowd, S. Puri,A., Graybiel, A. *Dopamine and Astrocytes: A Dynamic Duo Driving Behavioral Engagement or Exploration-Exploitation?* MIT BCS-MIBR-PILM poster presentation 2023 (4, June. 2023)
- **Ahn, G.**, Choi, B.S. Ko, S., Jo, C., Han, H.S., Lee, M.C. and Ro, D.H. *High-Resolution Knee X-ray Image Synthesis Using a Style Generative Adversarial Networks Adaptive Discriminator Augmentation (StyleGAN2-ADA)*. 65th Annual Congress of the Korean Orthopaedic Association 2021 (14-16, Oct. 2021)
- **Ahn, G.**, Choi, B.S., Ko, S., Jo, C., Han, H.S., Lee, M.C. and Ro, D.H. *Generating Images Showing Arthritis Progression by Latent Space Control Using Generative Adversarial Networks*. 2021 Korean Society for Cartilage and Osteoarthritis Summer Symposium (27, Aug. 2021)
- Kim, K., Joo, Y., **Ahn, G.**, Cha, J. *Sex, brains, and individuals: Machine-learned brain sex explains individual-specific cognitive development and its genetic factors in prepuberty children*. Society of Biological Psychiatry. (Apr 29 – May 01, 2021)
- **Ahn, G.**, Kim,B. and Cha, J. *Identifying Prepubertal Children with Risk for Suicides Using Deep Neural Network Trained on Multimodal Brain Imaging*. SfN Global Connectome. (11-13, Jan. 2021)
- **Ahn, G.** *Other Race Effect of Facial Expression Recognition through Machine Learning*. XAICON 2020 sponsored by the Center for Humane AI. (19, Jun. 2020)

## HONORS AND AWARDS

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- Henry E. Singleton (1940) Fellowship, MIT, awarded in consecutive years for graduate research in brain and cognitive sciences (Sep. 2024 - May 2026)
- Biomedical Global Talent Nurturing Program (\$40,000), for future Biomedical engineers who will lead national society, Korea Health Industry Development Institute (Jan. 2023-Dec. 2023)
- Certificate of Appreciation, Yoo Suk Hong, Dean, College of Engineering, SNU (Sep. 2022) : Appreciation for the contribution of youth scholarship to the education and research foundation
- Engineering Scholarship (\$300), for future engineers who will lead national society, Daesung foundation (Mar. 2022)
- Future Leadership Scholarship (\$4,000), for the future leader who will lead national society, Dojinhwa foundation (Sep. 2021, Mar. 2023)
- Certificate of Appreciation, Kookheon Cha, Dean, College of Engineering, SNU (Jul. 2021) : Acknowledgement of genuine and creative efforts at the forefront of AI education in Korea
- Academic Excellent Scholarship (\$12,000), High GPA, Ilsong Foundation (Mar. 2014 - Feb. 2020)
- Academic Excellent scholarship (\$3,000), High GPA, SNU (Aug. 2018, Feb. 2019)